

Shadman Salif Swanan

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EDUCATION

BRAC University

Jan. 2022 – Present

Bachelor of Science in Computer Science and Engineering (CSE)

Dhaka, Bangladesh

- **CGPA:** 3.82 out of 4.00
- **Programming Coursework:** Fundamentals of Programming, Object-Oriented Programming, Data Structures & Algorithms
- **Computational Coursework:** Natural Language Processing, Computer Network, Linear Algebra, Neural Networks, Digital Logic Design, VLSI Design, Discrete Mathematics
- **Programming Languages:** Python, Java, C/C++, JavaScript, TypeScript
- **Frameworks & Tools:** PyTorch, OpenCV, NumPy, Matplotlib, Scikit-learn, Git, OpenGL, MySQL, React, Next.js, Figma

PROJECTS

Token-Level POS Tagging using RNN Models [🔗](#) Implemented POS tagging using RNN, LSTM, GRU, and BiLSTM architectures

Disease Prediction using Machine Learning [🔗](#) Developed classification models for diabetes prediction using KNN, Decision Tree, and Naive Bayes

Shikkha.com [🔗](#) Developed an educational platform using JavaScript for digital learning support

3D Car Racing Game [🔗](#) Built a 3D racing game using OpenGL with enemy AI, shooting mechanics, weather effects, and camera modes

Automated Sanitizer and Mask Dispensing Robot [🔗](#) Embedded systems project integrating automated sanitizer dispensing and mask detection

RESEARCH EXPERIENCE

Undergraduate Thesis Research

Sept. 2025 – Present

BRAC University

Dhaka, Bangladesh

- Conducting research on **semantic drift in the Bengali language** using diachronic word embeddings and transformer-based representations
- Reviewed and analyzed prior literature on semantic change detection, historical embeddings, and contextual language models
- Developing a column detection model for scanned Bengali newspapers to improve document layout understanding
- Building an OCR pipeline to construct a clean and high-quality diachronic text corpus for downstream NLP tasks

Leaf Disease Research (TinyResNet)

April 2026

Independent Research Project

Dhaka, Bangladesh

- Developed a lightweight deep learning model (TinyResNet) for tomato leaf disease classification under real-world conditions
- Proposed **distortion-aware augmentation** (blur, noise, low-light) to improve model robustness
- Conducted comparative analysis with ResNet18 baseline, achieving **96.59% accuracy** and superior generalization
- Evaluated model performance using confusion matrix, precision, recall, and F1-score across multiple disease classes

EXPERIENCE

University of Tokyo (Matsuo-Iwasawa Laboratory)

Apr. 2026 – Present

GCI World Program (Global Consumer Intelligence)

- Participating in an international AI education program focused on data science and machine learning
- Engaging with global participants to develop practical and theoretical understanding of AI applications

Grameenphone Academy

Mar. 2026

UX Design Decision-Making with AI & Design System Thinking with AI

- Completed certifications focusing on AI-assisted user-centered design and structured design system thinking
- Explored how AI enhances product design, decision-making, and scalable system development

Notre Dame College Model United Nations

Conference Management Officer

- Managed responsibilities in the Conference Management Department during a large-scale academic event
- Demonstrated strong organizational skills, adaptability, and ability to work efficiently under pressure

Notre Dame Eco & Space Club

Oct. 2018

Organizing Team Member – 2nd National Space Camp

- Contributed to organizing a national-level space camp across School, College, and Open categories
- Coordinated logistics and supported event execution for participants and organizers

Notre Dame Art Club

2018

Volunteer – 2nd National Art Carnival & Exhibition

- Supported event operations and assisted in organizing a large-scale art exhibition

CO-CURRICULAR ACTIVITIES

Notre Dame Nature Study Club

Vice President, Department of Photography

- Led photography initiatives and coordinated visual documentation for club activities and events
- Managed team members and contributed to creative direction and event coverage

Ideal English Language Club

General Member

- Assisted in organizing language-focused programs, workshops, and internal club operations
- Contributed to communication management and event coordination

BRAC University Residential Semester

Stage Drama Course Participant

- Completed an intensive stage drama course emphasizing teamwork, communication, and performance skills

BRAC University Sports Activities

Football Achievements

- Champion & Top Scorer, Intra-Spring 2022 Football Tournament, BRAC University
- Runners-Up, Crusade 2026 (Semester Tournament)
- Runners-Up, Residential Semester (RS) Football Tournament
- Runners-Up, RS Cup